



Nurses' Skill Assessment in Endotracheal Suctioning Using Video-based Activity Recognition

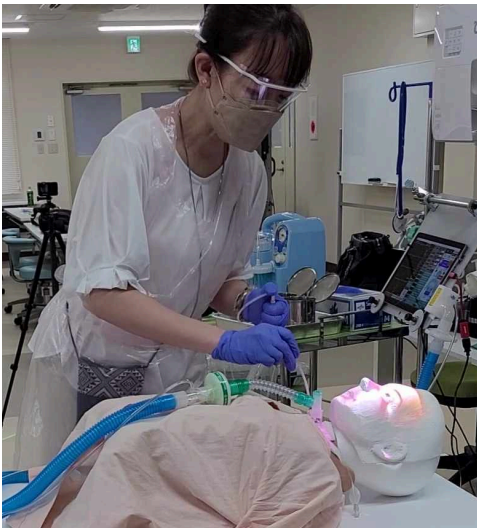
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Introduction

- **Endotracheal suctioning (ES)** is an **invasive** procedure.
- Skill assessment ensures nurses' competency for safe patient care, providing feedback for their skills.
- **Current assessment methods require staff** in real time. **Little work** for assessing nurses' skills automatically has been done in Endotracheal suctioning.



Executive Summary

Proposal



- ✓ Activity order
- ✓ Suction time
- ✓ Suctioning smoothness

- Propose an algorithm to define and track the main subject.
- Handle the complex activities.

Result

- Order score algorithm works correctly in ground truth.
- Recognized suction time is same ground truth with only 1 to 2 second difference.
- Suctioning smoothness result indicates that nurses perform ES more smoothly.
- Defining the main subject enhance accuracy by 3% and F1-score by 4%.
- Using Multiview data and training with micro labels improves accuracy by 6% and F1-score by 10%.

Conclusion

- Defining the main subject and smoothing the keypoints, combining multi-view pose extraction, post-processing, and skill assessment framework enhances activity recognition and understanding of nurses' performance.

Introduction

Challenges:

- Others included in the frame by passing in the background.
=> Propose an algorithm to define and track the main subject.
- Incomplete spatial and movement data, causing similarity in pose skeletons
=> Integrating data from multiple angles to capture pose skeleton variations
- Variability in complex ES activities confusing recognition
=> Segmenting these activities into shorter actions
- Pose skeleton similarities and misclassified predictions disrupting continuous long activity.
=> Introducing post-processing techniques



Related Work

Main Target Tracking

- Raj et al. [1] and Hur et al. [2] assigned person IDs for all subjects in a frame.
- Lack of automatically determining the main subject in each frame.

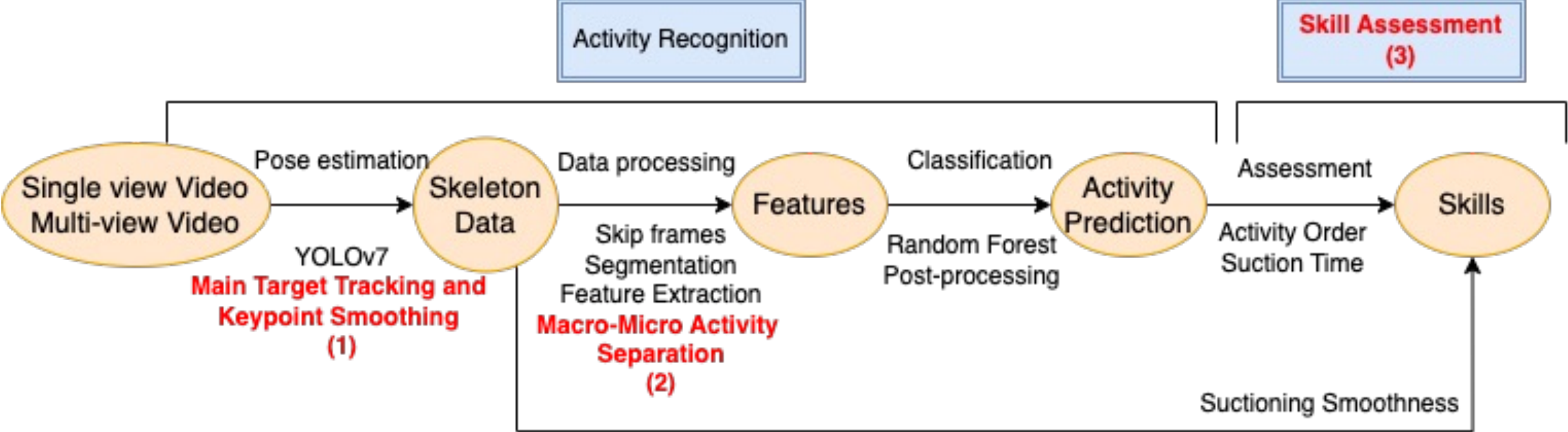
Macro-Micro Activity

- Faye et al. [3] analyzed human activities using both micro and macro approaches or Wieland et al. [4] predicted connections between micro-activity sequences.
- Lack of research in using micro activities for training and using macro activities for evaluation.

Skill Assessment

- Participants in Lim's study [5] need to watch recorded videos to reinforce skills
- Lack of assessment method without requiring direct observation.

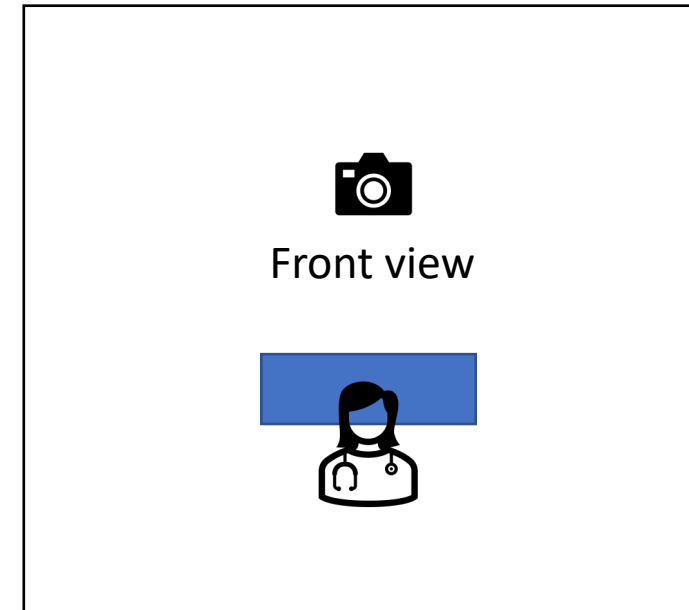
Flowchart



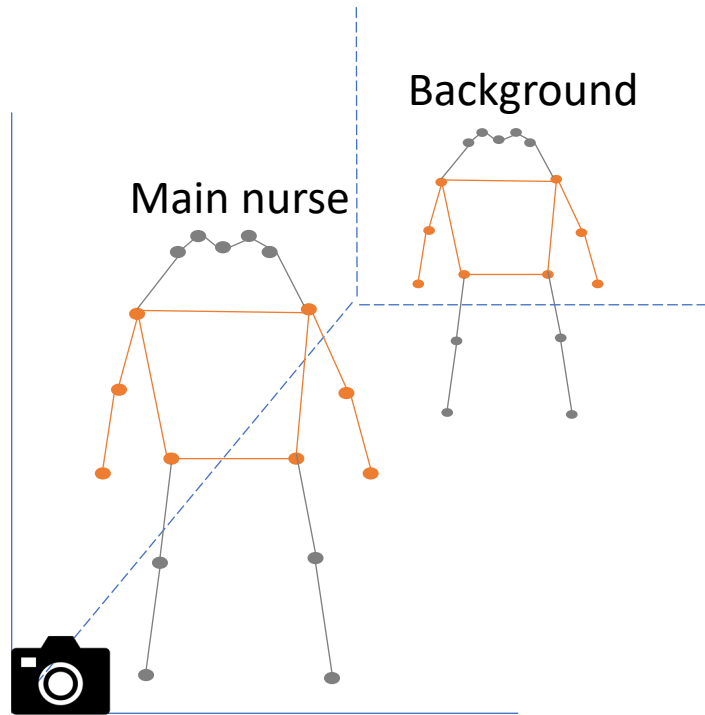
Single View Data

Participants: **10 nurses, 12 nursing students** from a university performing ES on the simulation system - ESTE-SIM.

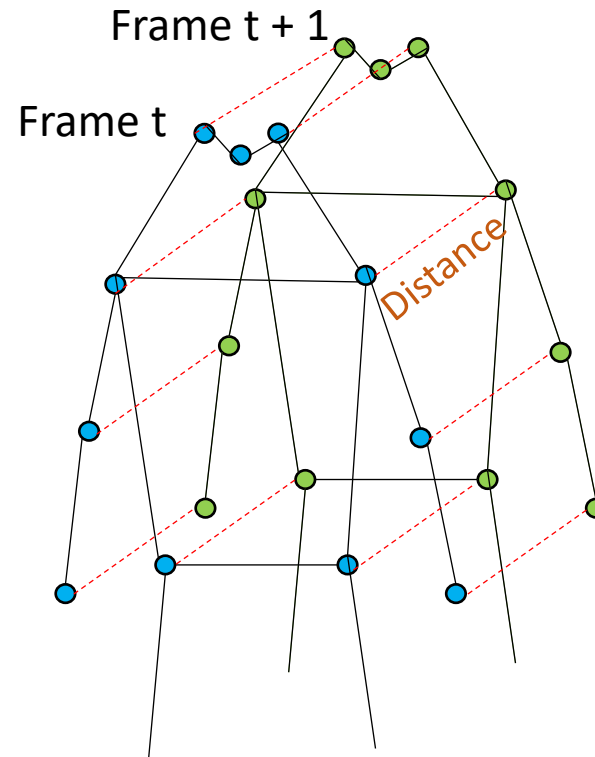
View: **Front** side of the nurse.



Main Target Tracking



Main nurse **near** the camera has a **greater** trunk length



Choose subject having **minimum** total distance



Keypoint Smoothing

Missing keypoints as (0,0) with a confidence score < 0.5 $\Rightarrow$ many short zero-value segments

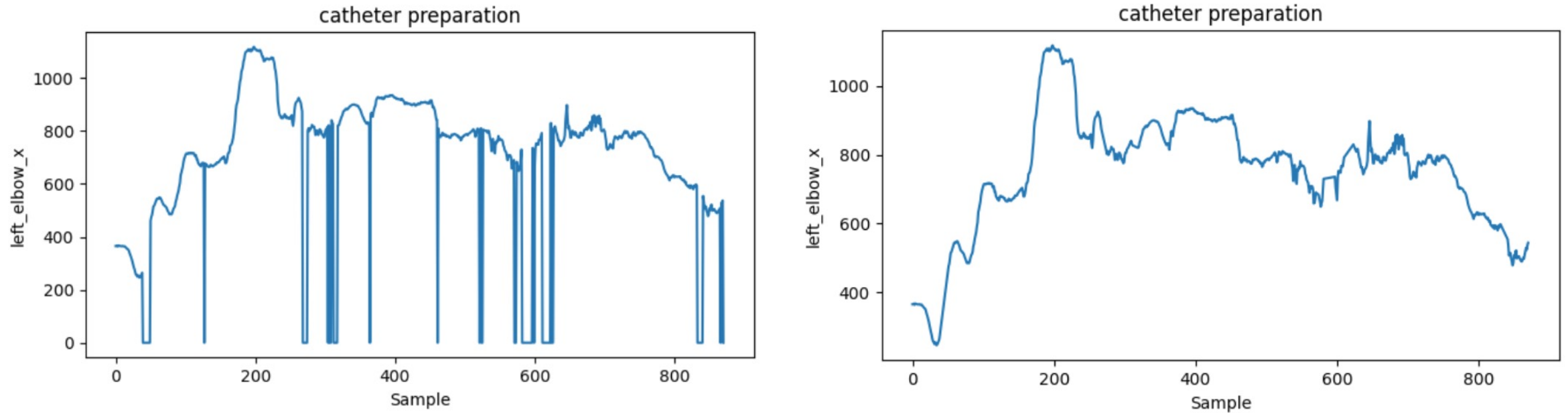


Figure: X coordinates of the left elbow in catheter preparation activity before smoothing (left) and after smoothing (right)

Main Target Tracking and Keypoint Smoothing

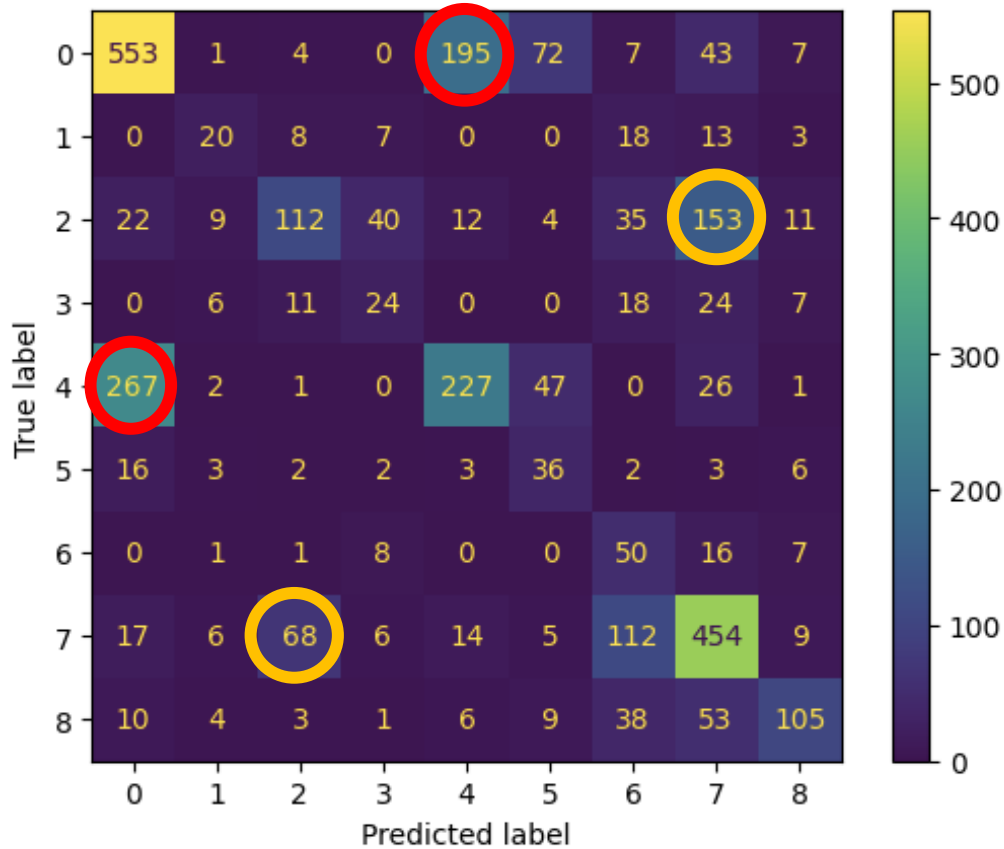
Experimental Overview

Approach	Pose Estimation	Feature Extraction	Data Segmentation	Algorithm
Baseline	After using YOLOv7, we chose randomly one subject to obtain keypoints	- Remove keypoints in knee and ankle. - Joint angles - Velocity	Window size: 2 s Overlapping rate: 50%	Oversampling: Synthetic Minority Oversampling Technique (SMOTE)
Our proposed	After using YOLOv7, we applied our algorithm and smoothing approach.			Random Forest

Main Target Tracking and Keypoint Smoothing

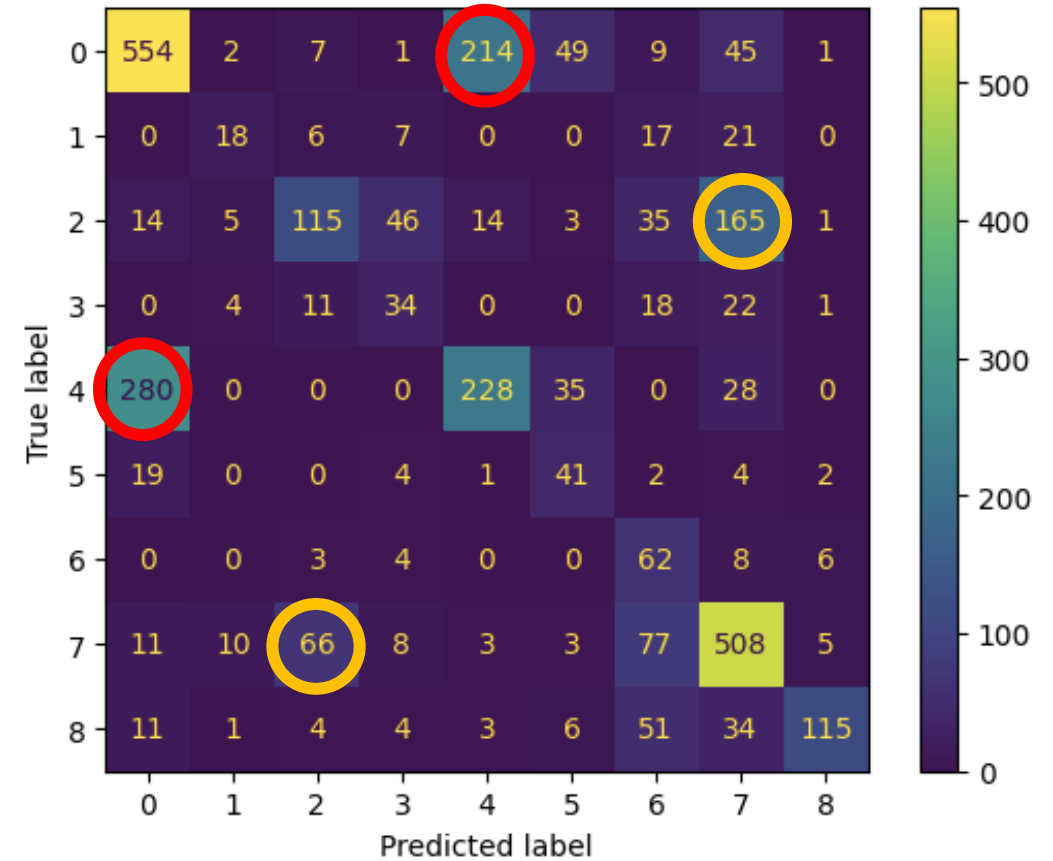
Result and Discussion

Approach 1 – Acc: 51% - F1-score: 42%



(a)

Approach 2 – Acc: 54% - F1-score: 46%



(b)

Figure: Confusion matrix of **whole testing data** by **Approach 1 – baseline approach** (a) and **Approach 2 – proposed method** (b)

Main Target Tracking and Keypoint Smoothing

Result and Discussion

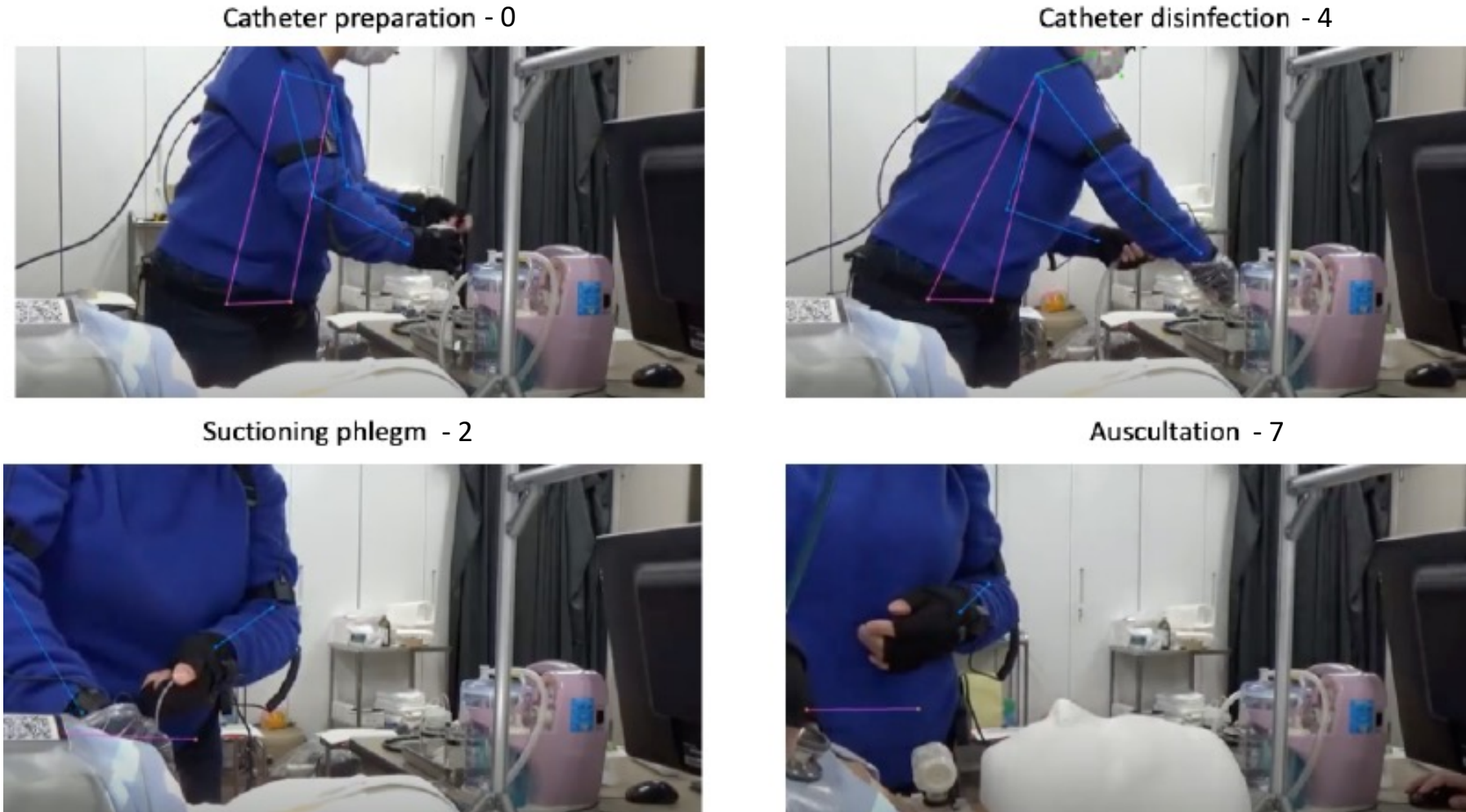
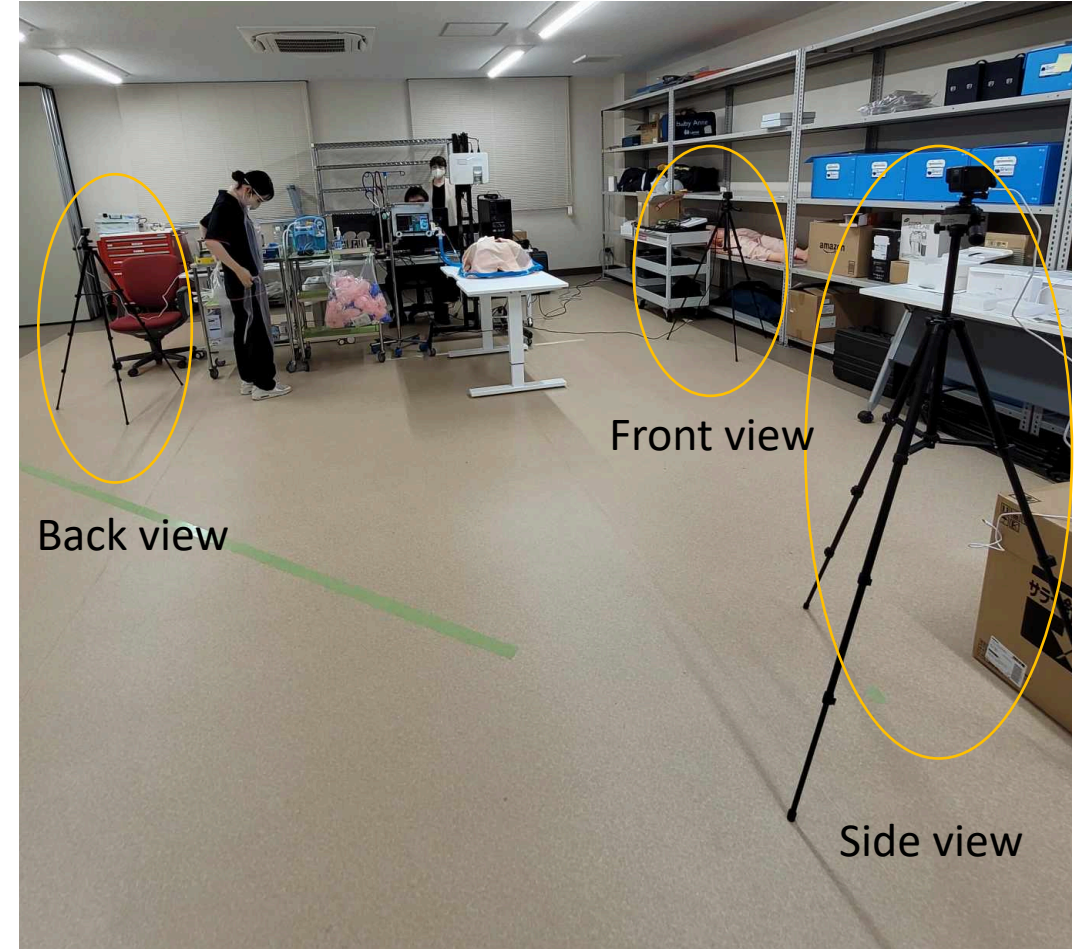
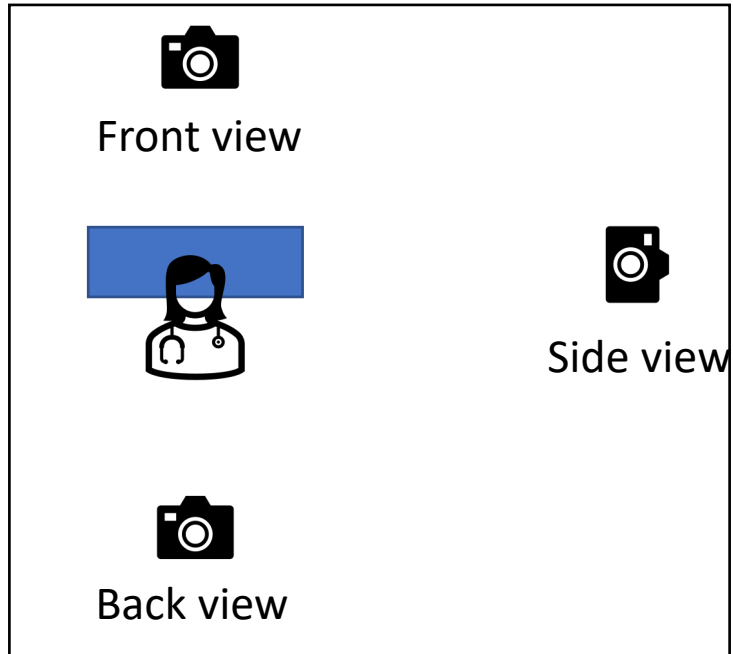


Figure: The similarity in skeleton between catheter preparation with catheter disinfection and suctioning phlegm with auscultation

Multi-View Integration and Macro-Micro Activity Separation

Participants: **7 nurses, 8 nursing students.**

View: Front, Side and Back view



Multi-View Integration and Macro-Micro Activity Separation

Activity name	Activity ID
Explanation to patient	0
Auscultation	1
Washing hands	2
Checking cuff pressure	3
Wearing PPE	4
Catheter preparation	5
Pressing button	6
Removal of airway	7
Suctioning	8
Refitting the airway	9
Wiping catheter	10
Washing catheter	11
Catheter disinfection	12
Removal of PPE	13
Others	14

Activity name	Activity ID
Explanation to patient	0
Auscultation	1
Washing hands	2
Checking cuff pressure	3
Wearing apron	4
Wearing goggles	5
Wearing gloves	6
Opening lids	7
Preparing alcohol tissue	8
Connecting suctioning catheter	9
Pressing button	10
Removal of airway	11
Suctioning	12
Refitting the airway	13
Wiping catheter	14
Washing catheter	15
Disconnecting catheter	16
Closing lids	17
Putting back catheter	18
Removal of gloves	19
Removal of goggles	20
Removal of apron	21
Others	22

Multi-View Integration and Macro-Micro Activity Separation

Experimental Overview

Approach	Data	Data processing	Feature Extraction	Algorithm
Baseline	- Front view - Macro labels (15 activities) - Without skip frame	- After using YOLOv7, we applied Main Target Tracking and Keypoint Smoothing. - Window size: 2 seconds - Overlapping rate: 50%	- Joint angles - Velocity of each frame. - Slow motion - Bone length - Timestamp.	Oversampling method: SMOTE
1	- Front view - Macro labels (15 activities)			Random Forest
2	- Front + Back view - Macro labels (15 activities)			
3	- Front + Back view - Micro labels (23 activities)			

Multi-View Integration and Macro-Micro Activity Separation

Result Analysis – Multi View Data

Approach	Data	Accuracy		F1-score	
		Micro label (23 activities)	Macro label (15 activities)	Micro activities (23 activities)	Macro activities (15 activities)
Baseline	- Front view - Training with Macro labels (15 activities) - Without skip frame		71%		51%
1	- Front view - Training with Macro labels (15 activities)		73%		52%
2	- Front + Back view - Training with Macro labels (15 activities)	–	74%	–	51%
3	- Front + Back view - Training with Micro labels (23 activities)	74%	77%	58%	61%

Post-processing

Misclassified cases:

7. Opening lids => 6. Wearing gloves, 9.

Connecting catheter

13. **Refitting** airway => 11. **Removal** of airway

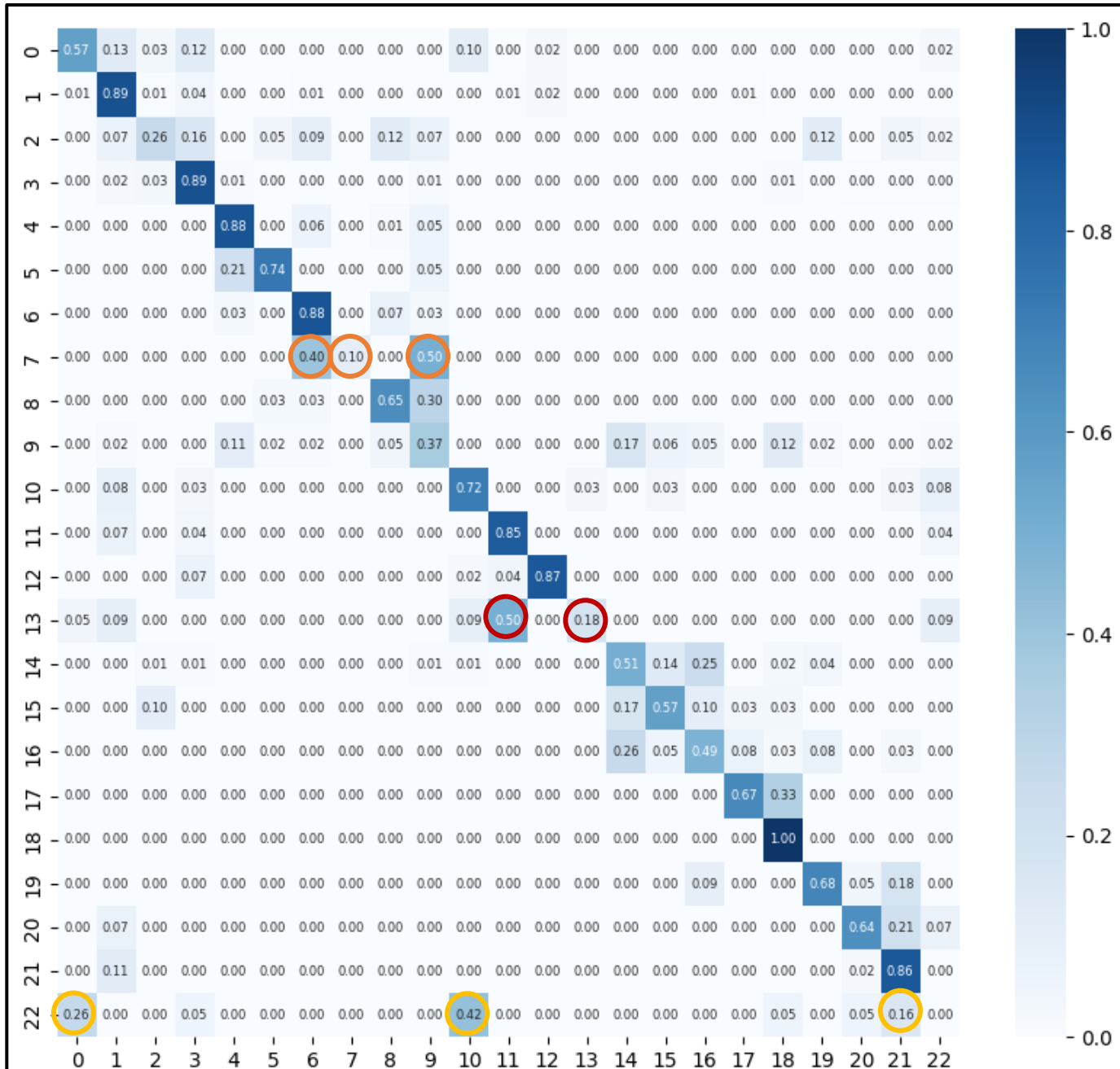
22. Others => 10. pressing button



Removal of airway

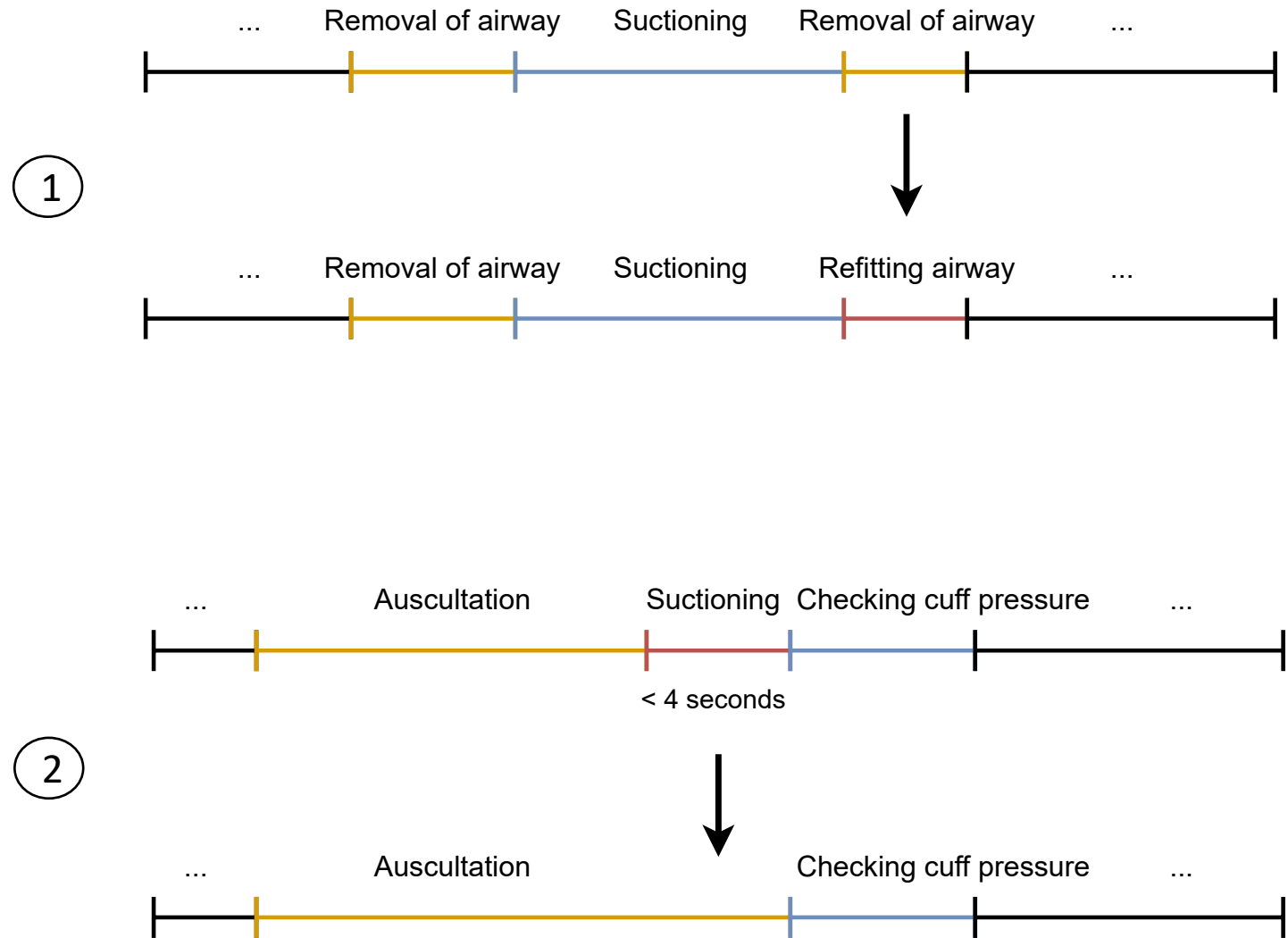


Refitting airway

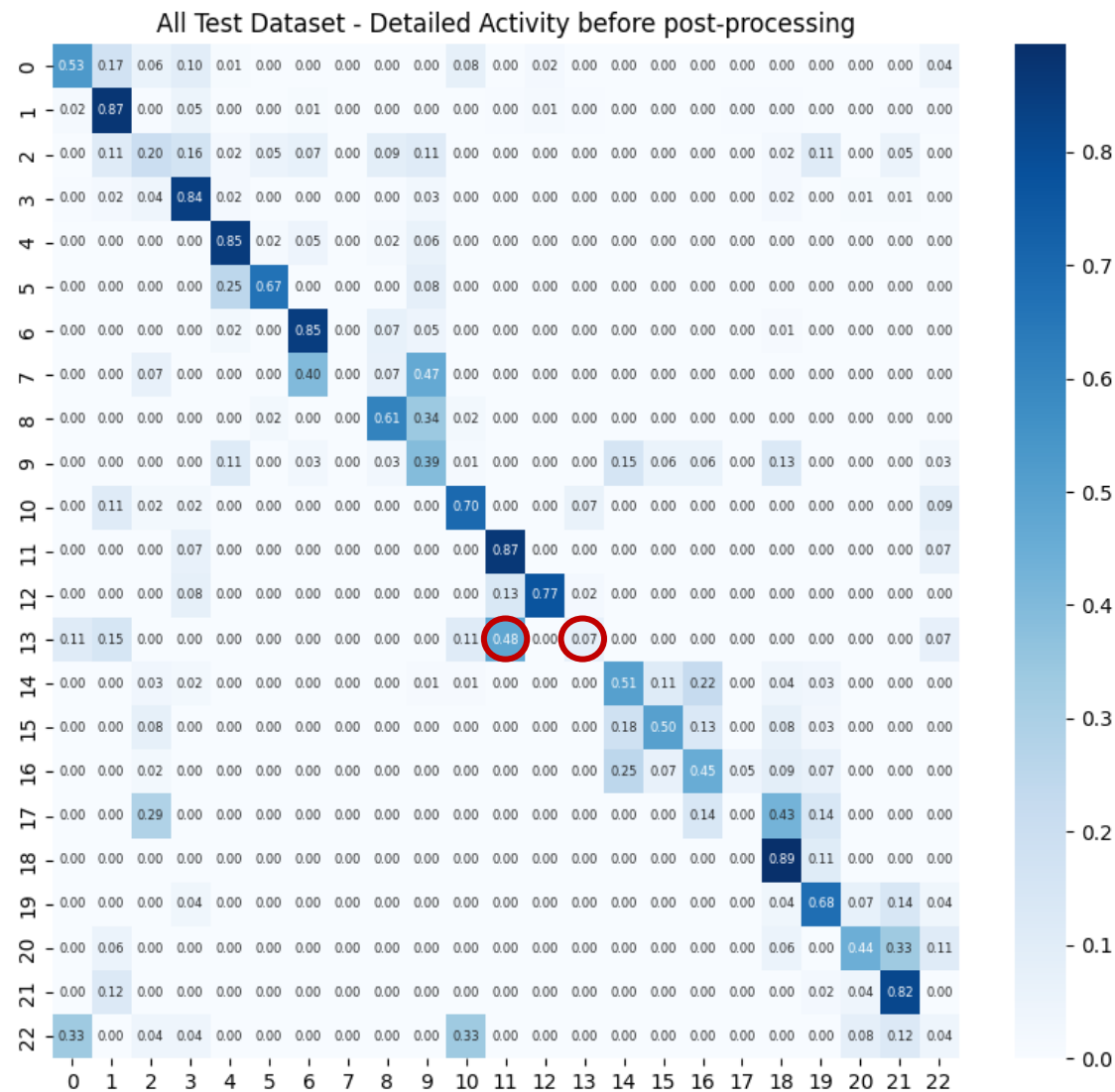


Post-processing algorithm

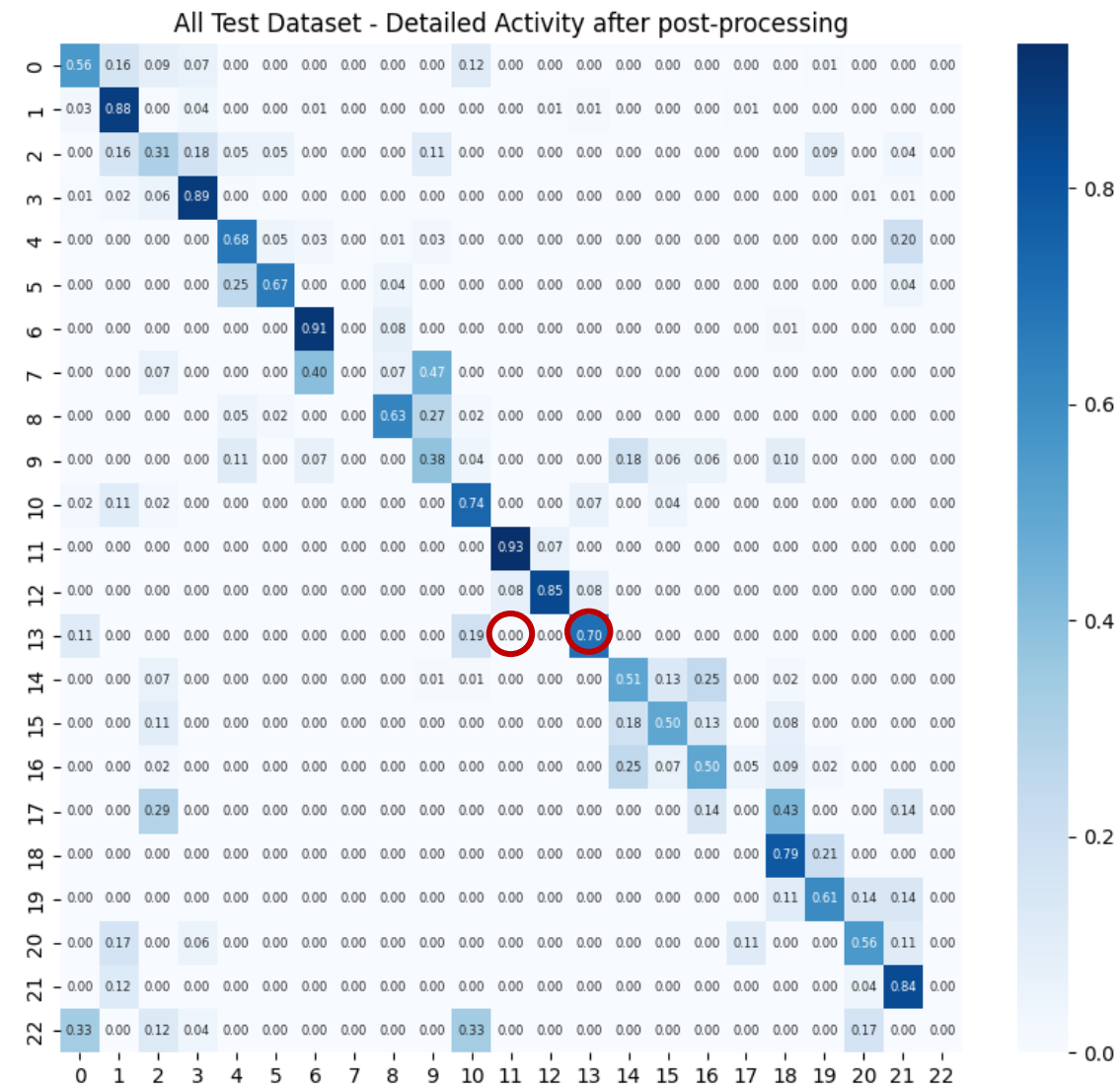
- Removal of airway vs Refitting airway
- Merge short segments (< 4 seconds)



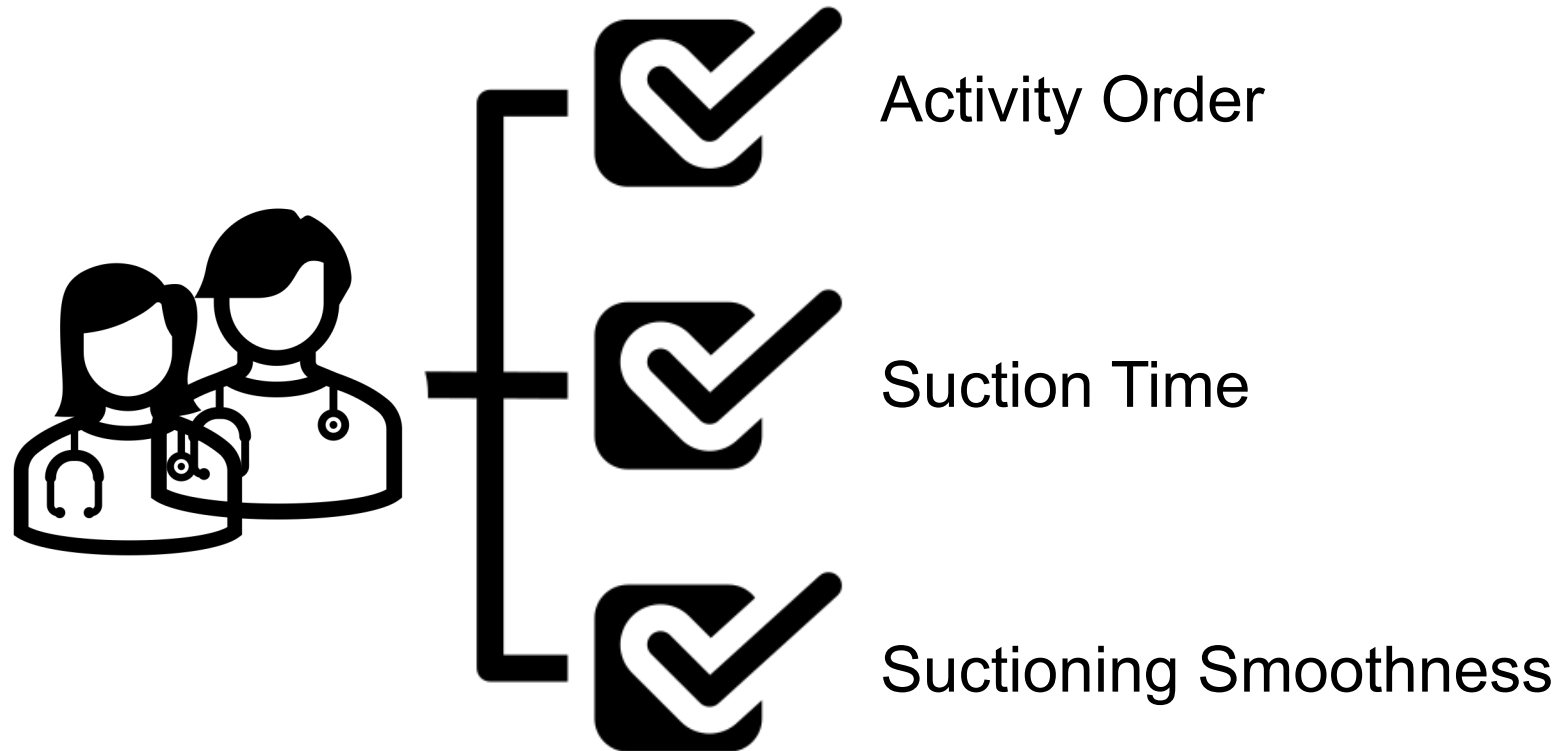
Post-processing



Acc: 69%
F1-score: 52%



Acc: 71%
F1-score: 56%

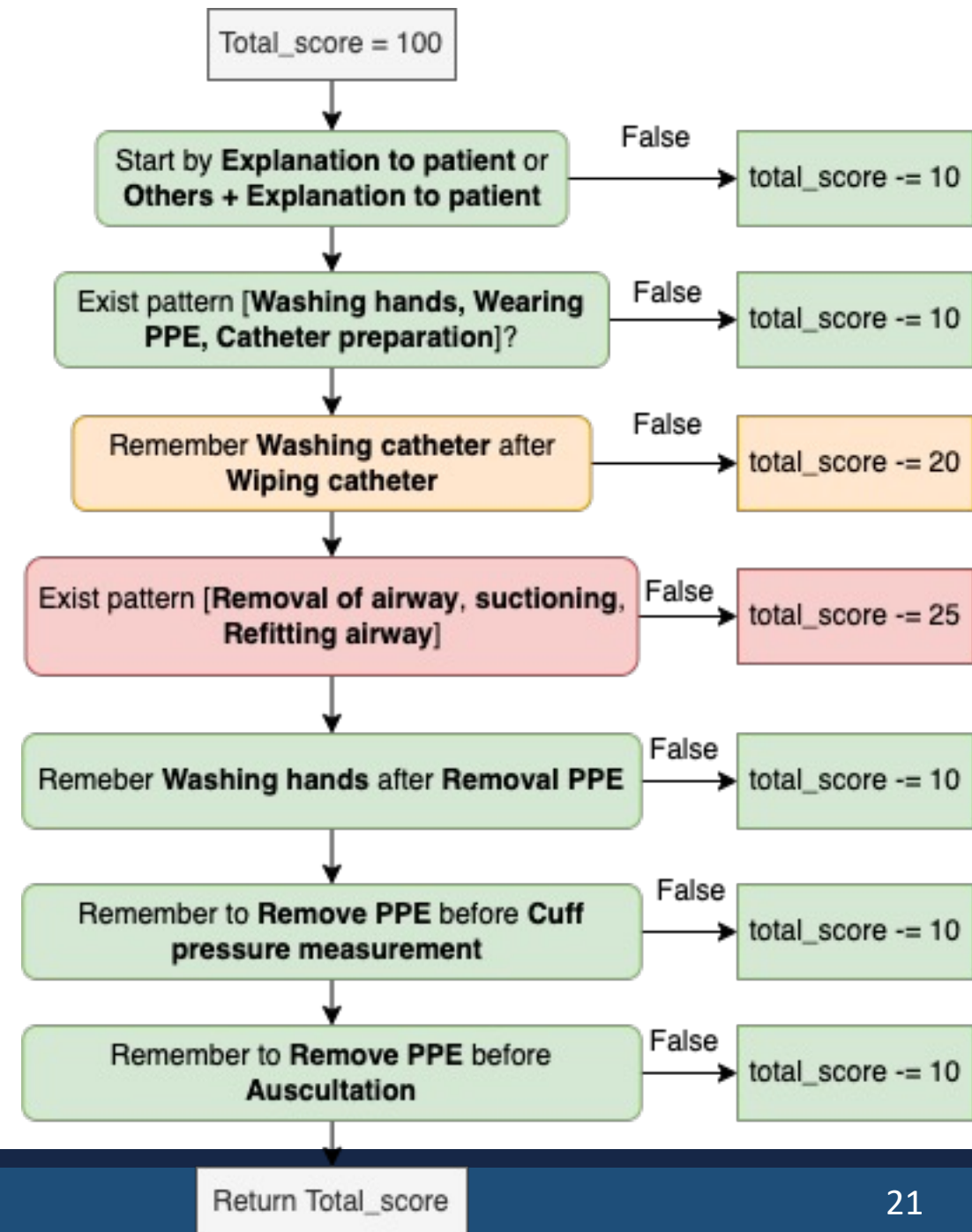
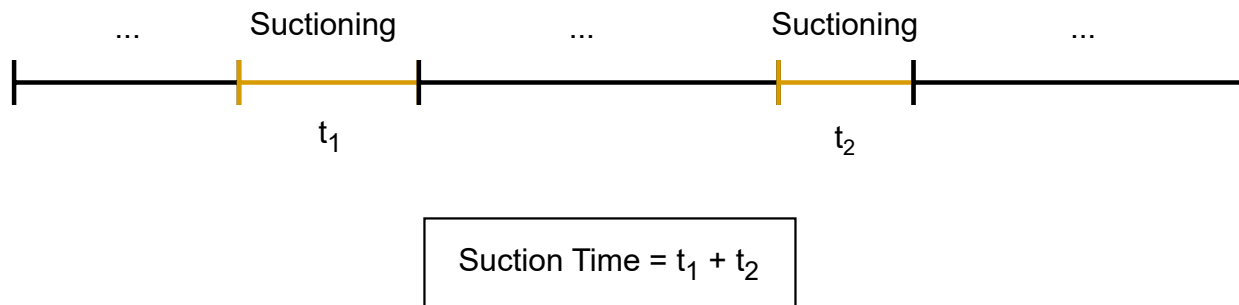


Skill Assessment

Activity Order

- Initially, the score is set as 100.
- Depending on the importance of each mistake that nurses made, we will minus their score with a corresponding value.

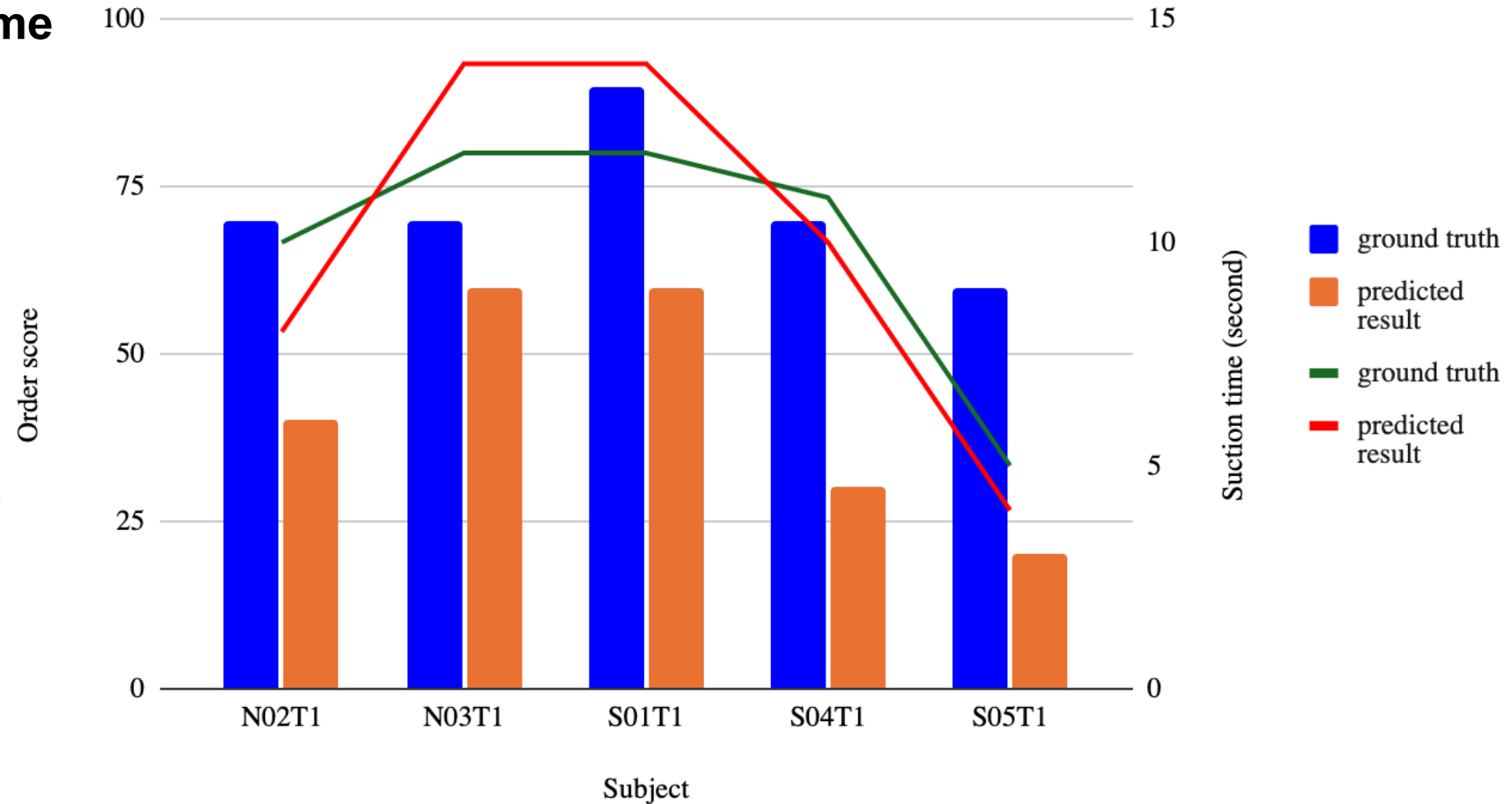
Suction Time



Skill Assessment

- **Activity order and Suction time**

- The suction time determined from predicted results has a similar trend as the suction time from ground truth.
- However, the order scores from predicted results are always lower than the ones from ground truth.

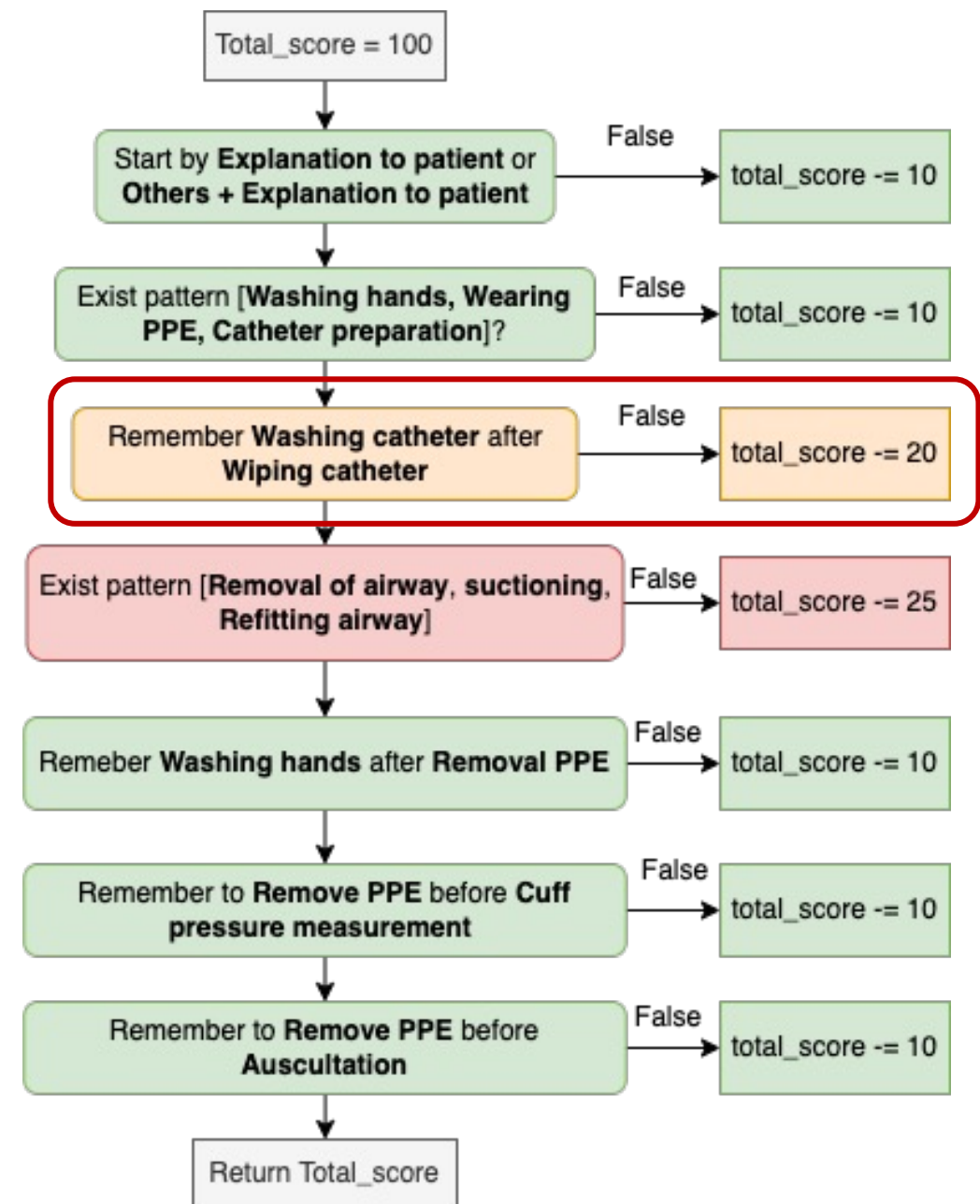


Skill Assessment

- Activity order and Suction time

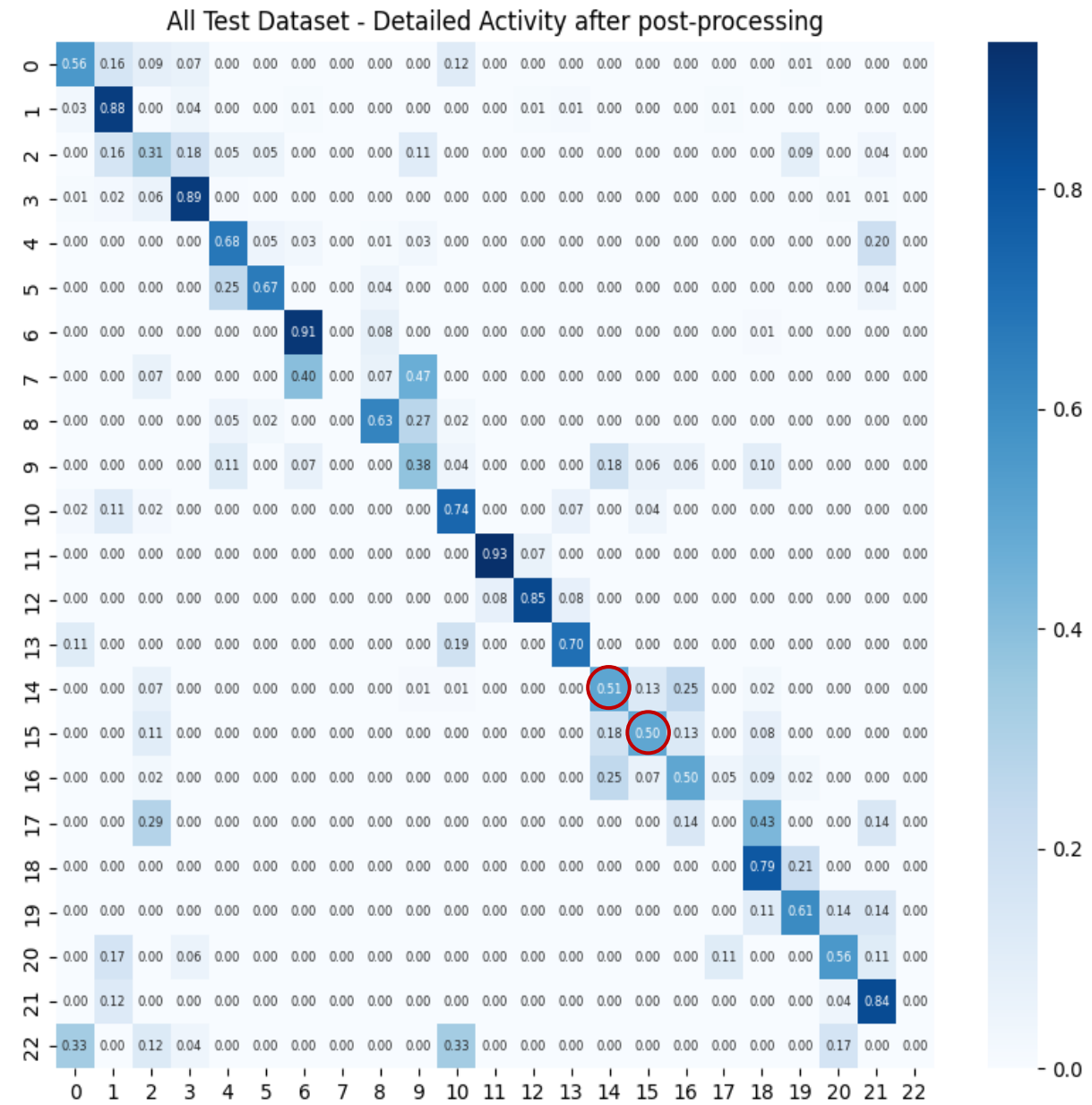
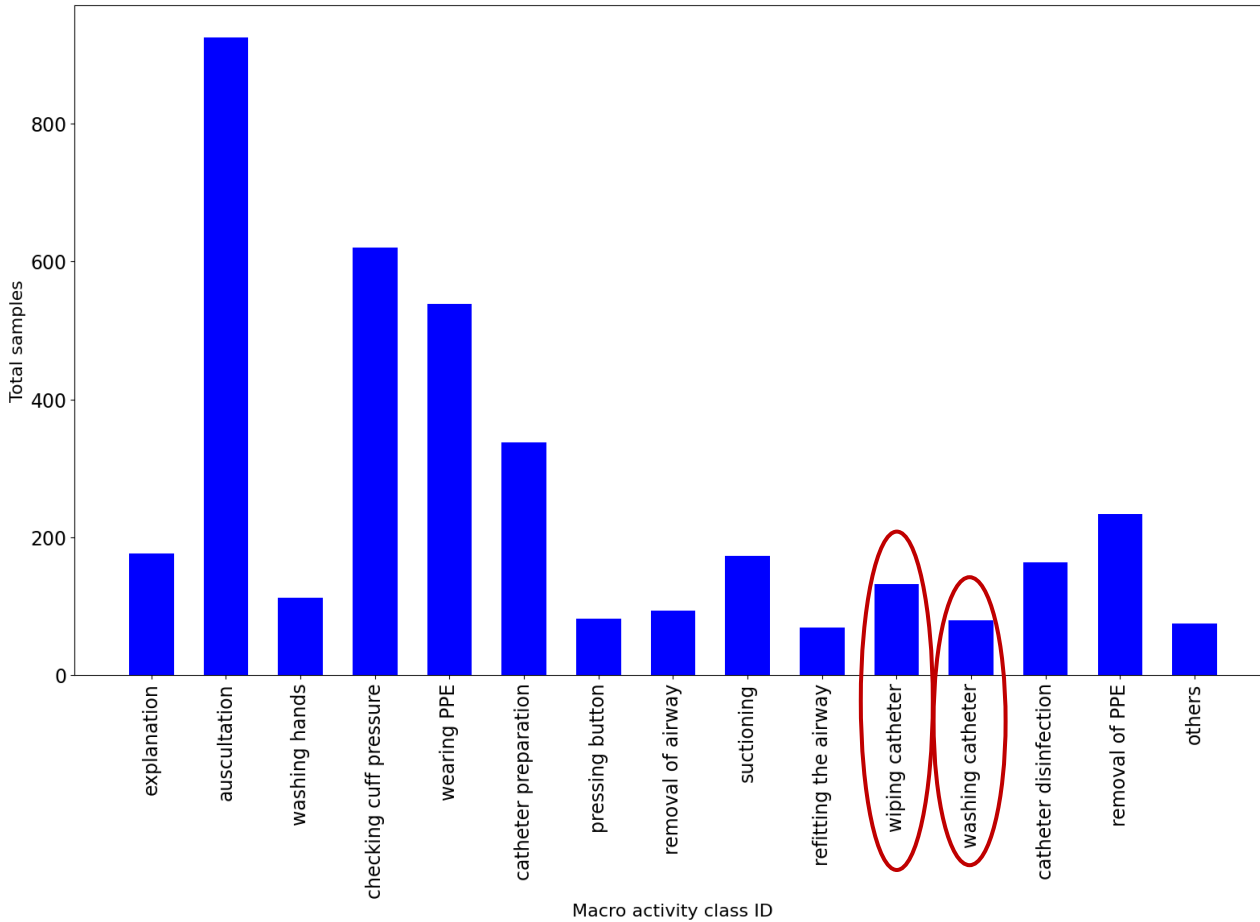
Kind of mistake	Incorrect times
Forgetting to wash catheter after wiping it	5
Not existing [Washing hands, Wearing PPE, Catheter preparation]	3
Not remembering to remove PPE before checking cuff pressure	2
Forgetting to wash hands after removal of PPE	1
Not remembering to remove PPE before auscultation	0

- "Forgetting to wash catheter after wiping catheter" mistake has the highest incorrectly recognized frequency and a high penalty with a minus 20 score.



Skill Assessment

- Activity order and Suction time



Suctioning Smoothness

Joint Angle Velocity

- How rapidly joint angles change over time
- Lower value -> smoother action

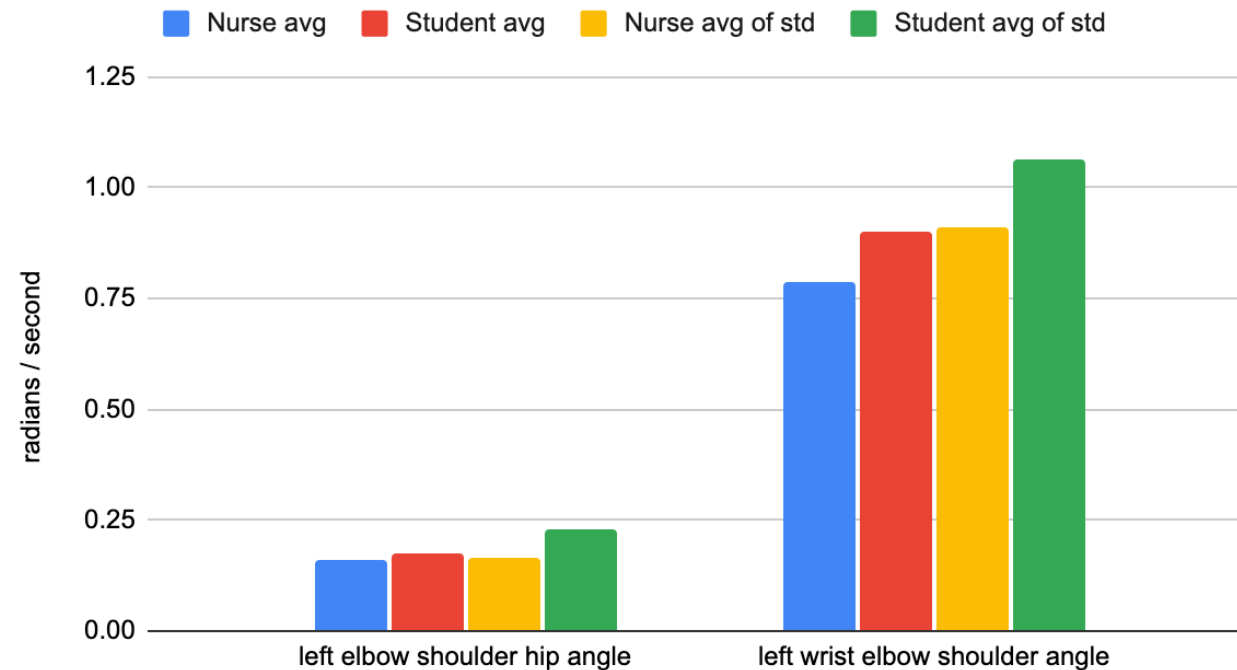
Total Variation

- The trajectories of specific joints during the suctioning procedure
- Lower value -> smoother action

Skill Assessment

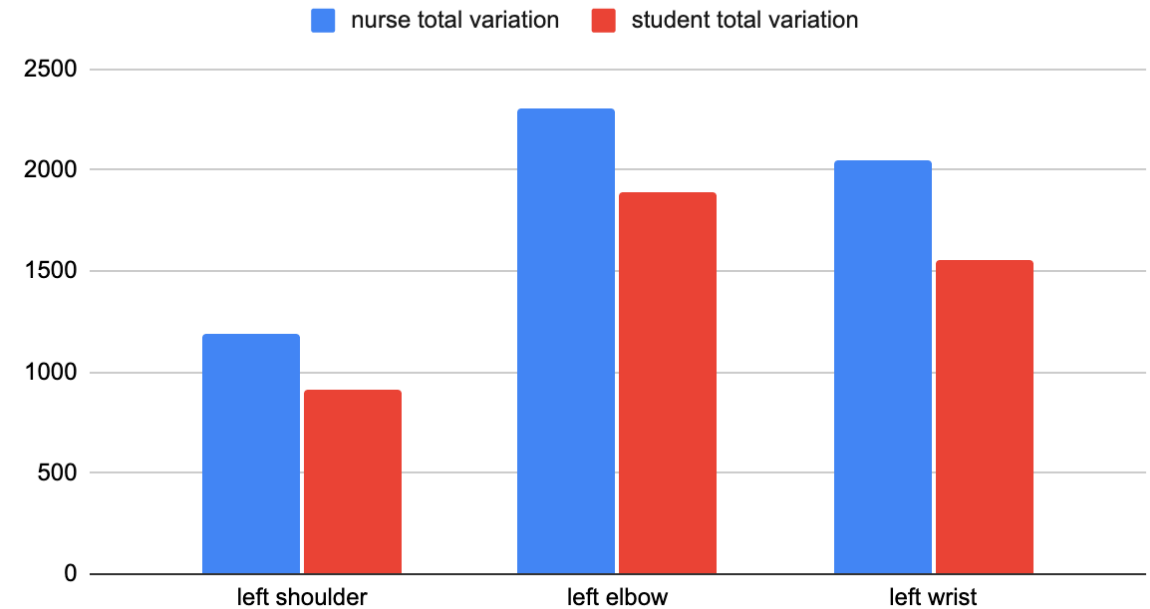
- Suctioning smoothness

Joint angle velocity between Nurses and Students



The std and average amplitude in velocity of nurses are smaller than students in both angles.

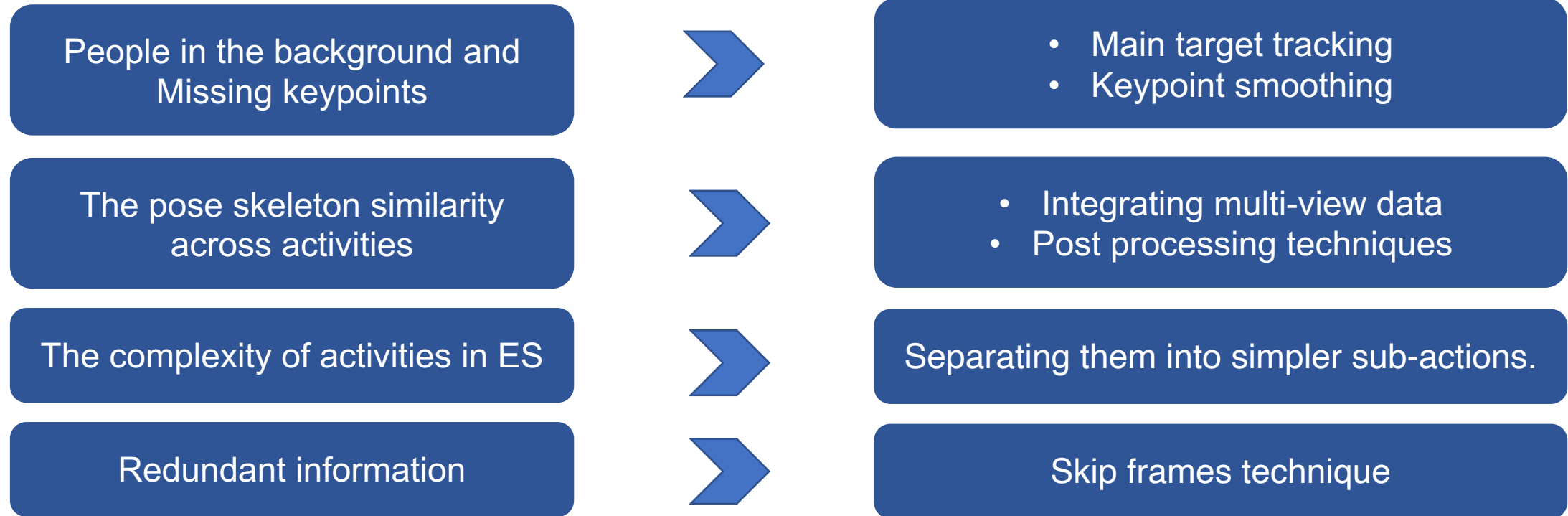
Total variation of nurse and student



However, the variation in skeleton of nurses when performing suctioning is higher than students.

Conclusion

- **Automatic** nurses' skills **assessment** in ES based on **the order, duration, and smoothness of actions**.
- Enhance performance of activity recognition systems by:



- The model's performance in predicting important activities affects the order score difference between ground truth and prediction.
- Future work, we intend to explore alternative data features or representations to better distinguish important activities.

References

- [1] Bharath Raj N., Anand Subramanian, Kashyap Ravichandran, and N. Venkateswaran. Exploring techniques to improve activity recognition using human pose skeletons. In 2020 IEEE Winter Applications of Computer Vision Workshops (WACVW), 2020.
- [2] Paul Hur and Nigel Bosch. Tracking individuals in classroom videos via postprocessing openpose data. USA, 03 2022.
- [3] Sebastien Faye, Nicolas Louveton, Gabriela Gheorghe, and Thomas Engel. Toward a characterization of human activities using smart devices: A micro/macro approach. In 2016 IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS), pages 676–681, 2016.
- [4] Christoph Wieland and Victor Pankratius. Detection and validation of macroactivities in human inertial signals using graph link prediction. *Sensors*, 24(4), 2024.
- [5] Lim Kyung Choon. Simulation-based clinical judgment and performance ability for tracheal suction in nursing students. *J Korean Acad Soc Nurs Educ*, 23(3):330–340, 2017.

THANK YOU